openvaf-r — Full Change Report

Every modification and its reason (original $\rightarrow$ current)

Ngspice + OpenVAF Enhancements project

July 2026

Contents

openvaf-r — Full Change Report

Every modification applied to the OpenVAF-Reloaded compiler in this project, with its reason. The baseline is the pristine OpenVAF-Reloaded source tree (as vendored in the project's `original/` snapshot); the current state is the tree committed in this repository under `OpenVAF-master-20260610/`. Each entry links the enhancement write-up in `enhancements_doc/` that carries the full engineering detail and the verifying example suite. The companion document ngspice_changes_full-report.md covers the simulator side.

> Maintenance note: this report is updated whenever an enhancement touches compiler sources. The per-enhancement index at the end tells you the last change it covers.

Scope summary. ~120 modified source files across the compiler pipeline, two new files (`hir/src/elaborate.rs` — the elaboration pass — and `hir_def/src/item_tree/diagnostics.rs`), plus three classes handled wholesale: the `test_data/` fixtures and snapshots (95 files — new test inputs and snapshots regenerated whenever a feature legitimately changed descriptor contents), the `integration_tests/*/*.osdi` reference objects (regenerated at each ABI bump), and the removal of the AGPL `vacask` submodule. The compiler builds warning-free; every change below was regression-gated by the example-suite arsenal, the integration suite, and the VA_TEST industry-model corpus.

---

1. Lexer — **`openvaf/lexer/`** (**`lib.rs`**, **`cursor.rs`**)

- Based integer literals (`8'hFF`, `'sb101`): the lexer had only a commented sketch; underscores crashed it. Both the sized and unsized forms now lex, requiring a digit after the base (a silent-0 trap otherwise) (E-46).
- Don't-care digits `x/X/z/Z/?` accepted in the binary/octal/hex digit runs for `casex`/`casez` items (E-78).
- Escaped identifiers (`\my-net!`) lexed per the LRM (E-46).
- **`// comment at EOF`** hang fixed: with no trailing newline the line-comment loop spun forever on the EOF character — the only EOF-unsafe loop in the lexer (E-35).

2. Tokens and grammar — **`openvaf/tokens/`**, **`openvaf/parser/`**, **`veriloga.ungram`**, **`sourcegen/`**

`tokens/src/parser/generated.rs` gains the keywords and node kinds each language feature needed, threaded through `parser/src/grammar/*` (`stmts.rs`, `expressions.rs`, `items.rs`, `items/module.rs`), the grammar source of truth `syntax/veriloga.ungram`, and the sourcegen/ generators (`ast/src.rs`, `hir_builtins.rs`, `mir_instructions.rs` — the templates the generated token/AST/builtin/instruction tables come from):

- `do … while` post-test loops (E-19);
- `paramset`/`endparamset` blocks (E-21);
- `{…}` concatenation and `{n{…}}` replication as real operators, with `'{…}` remaining the typed aggregate (E-34);
- array-literal expressions actually parsed (they were fully scaffolded but unreachable) (E-4, E-14);
- vectored (bus) net declarations and bit-selects (E-3), multi-index bit-selects for N-D arrays (E-15), and the LRM name-then-range declaration order (E-18);
- module instantiation items (E-5), `generate for`/`genvar` (E-8), and `generate if`/`generate case` with optional block labels (previously mis-parsed into a broken `GENERATE_FOR`; E-67);
- `defparam` — a reserved word with no grammar rule (E-58);
- event OR lists `@(cross(…) or timer(…))` via a new `or` keyword and looped event grammar (E-59);
- `casex`/`casez` sharing the case rule, the keyword token carrying the flavor (E-78);

- `@(initial_step)`/`@(final_step)` with analysis-phase lists (E-7, E-53);
- indirect branch assignment `V(x): V(y) == expr;` (E-2);
- the <<</>>> arithmetic-shift tokens (plus a pre-existing lexer bug fix) (E-6);
- panic-free direction parsing: `port_decl`/`func_arg` asserted the next token was a direction (`bump_ts`), which any non-Verilog text reaching a module-head port list turned into a compiler crash; both now emit a diagnostic with one token of forced progress (E-84);
- operator precedence corrected against LRM Table 4-2: `%` bound tighter than `*`/`/` (so `6*7%4` evaluated to 18, not 2), and xnor split from xor (E-38);
- derived-nature parents emitted the wrong node kind (`NAME_REF` where the AST wanted `Path`), leaving the entire inheritance machinery unreachable (E-39).

3. Preprocessor — **`openvaf/preprocessor/`** (+ **`basedb`** diagnostics)

- Ten compiler directives that previously hard-failed as undefined macros: `` `default_discipline ``, `` `celldefine ``/`` `endcelldefine ``, `` `unconnected_drive ``/`` `nounconnected_drive ``, `` `timescale ``, `` `line ``, `` `pragma ``, `` `undefineall ``, `` `default_nettype `` (E-6).
- **`` `default_transition ``** parsed and threaded to `transition()` lowering (E-47).
- Recursive macro expansion crashed the compiler (stack overflow, direct and mutual): the `MacroRecursion` diagnostic existed but was never emitted, and its report renderer was a `todo!()`. The guard pushes around the macro *body* expansion only — an entry-scoped guard false-positives on the legal `QUAD(x) = TWICE(TWICE(x))` nesting (E-65).

4. Syntax / AST — **`openvaf/syntax/`** (**`ast.rs`**, **`generated/nodes.rs`**, **`expr_ext.rs`**, **`node_ext.rs`**, **`name.rs`**, **`validation.rs`**, **`error.rs`**)

- Typed AST nodes for every new construct above (generated from the ungrammar).
- Literal decoding (`expr_ext.rs`): based-int parsing with size/sign handling (E-46) and the mask-aware variant returning `(value, x_mask, z_mask)` for casex/casez (E-78); a compiler crash on large bare-integer-shaped literals fixed by falling back to a float constant (E-4).
- String unescaping rewritten as a single pass: the sequential `str::replace` chain corrupted overlapping escapes (`\\n`) and octal `\ddd` was missing; all consumers route through one function (E-48).
- **`Name::resolve`** ate the last character of escaped identifiers (the committed `std.va` snapshot had baked the bug in as `logi`) (E-46).

5. Elaboration — **`openvaf/hir/src/elaborate.rs`** (new file)

The compile-time flattening pass: every stage downstream of the parser is architected around exactly one flat module per artifact, so hierarchy is resolved by recursively inlining instantiated modules — alpha-renamed per instance, ports bound to the caller's nets, parameters bound to overrides — into an ordinary flat module *before* the rest of the pipeline runs. Built for module instantiation (E-5) and grown into the home of every structural feature:

- `generate for`/`genvar` unrolling (E-8), nested loops, anonymous blocks, `generate if`/`case` with elaboration- time constant folding, and the genvar-substitution fix (genvars in ordinary expressions were routed through identifier renaming, which re-escaped `1e3*(i+1)` into a broken escaped identifier; they now become literal-value holes) (E-67);
- implicit nets from undeclared instance connections, discipline derived from the connected port, prefixed per instance so no cross-instance shorts (E-41);
- hierarchical names `u1.u2.node` and `$root` via an instance-chain map and a hole scanner (E-49);
- **`defparam`** resolved through the same chain rewrite, with the LRM-mandated precedence over instance `#()` overrides (E-58);
- net concatenation in port connections expanded bit-by-bit onto vectored ports (E-59);

- bus slicing onto instance arrays and bus ports (E-5); paramset twin-module rendering (E-21); net initializers preserved through re-rendering (E-45); escaped identifiers re-rendered correctly (E-46);
- undefined instantiation targets are a hard error — they used to be silently dropped from the rendered output, turning a typo'd module name into an invisible open circuit; discipline-named misparses (`electrical out[0:2];` parses as an instantiation) and paramsets dropped over unresolvable targets get tailored messages (E-84);
- **`$port_connected`** resolved at flattening time to a literal `(1)`/`(0)` per instance — after inlining, an open port is just a synthesized local net, so the builtin used to fail validation in exactly the unconnected case it exists to detect; top-level modules keep the native OSDI connected-flag path (E-84).

6. Definition layer — **`openvaf/hir_def/`** (+ **`openvaf/hir/`** mirrors)

`body.rs`/`body/lower.rs`/`body/pretty.rs`, `expr.rs`, `item_tree*`, `nameres*`, `builtin.rs`, `data.rs`, `types.rs`, `db.rs`, plus the new `item_tree/diagnostics.rs` and the `hir` façade (`body.rs`, `lib.rs`, `db.rs`, `diagnostics.rs`):

- statement/expression collection for every new construct: `do…while` (E-19), `repeat` and `disable` (E-9), event OR lists (E-59), concat/ replication (E-34), array aggregates and whole-array assignment (E-14), `CaseKind` + per-item don't-care masks with a stray-literal list (E-78);
- N-D array machinery: per-dimension size lists, multi-index bit-selects, per-element parameter expansion (E-15); declaration initializers on (N-D) arrays via a `Var::array_index` leaf split (E-43);
- **`localparam`** made genuinely non-overridable (it behaved exactly like `parameter`) with derived localparams still tracking inputs (E-9); paramset lowering as twin modules whose bound params are localparams with override expressions (E-21);
- name resolution: `electrical ground gnd;` ordering (E-9), nature-attribute access (`net.potential.abstol` — the third "scaffolded-but-unwired resolver boundary" found) (E-45), derived-nature resolution (E-39);
- **`builtin.rs`** (the system-function registry): the noise-table pair (E-9), the full `$random`/`$dist_*`/`$rdist_*` family (E-10), file I/O and string functions (E-11), the last fallback group (`$simprobe`, aliases, plusargs) — after which `is_unsupported()` is empty (E-12), `$simparam$str` (E-25), `$realtime` (E-59), `$table_model` (E-16);
- new diagnostics file: wrong-arity calls produced invalid HIR and crashed downstream — now a proper diagnostic (functions and parameters both) (E-43);
- multiple `analog` blocks collect into `entry_stmts` in source order — the as-if-concatenated LRM semantics, by construction (E-60).

7. Types and validation — **`openvaf/hir_ty/`**

`inference.rs` (+ `inference/fmt_parser.rs`), `builtin.rs` + `builtin/generated.rs` (signature tables), `lower.rs`, `types.rs`, `validation.rs`, `validation/body.rs`, `diagnostics.rs`:

- signature tables — a recurring defect source: `ANALYSIS` made variadic (E-30); `$table_model` given shape-synthesized varargs signatures for any dimension (a generic-varargs fallthrough *truncated* multi-arity signature lists — varargs builtins with signature lists need their own inference arm) (E-40); `TRANSITION`'s table was one argument short (3-argument calls crashed, 4-argument worked by accident) (E-47); `transition()` input typed Real, a `$dist` typo fixed (E-49); `SIMPARAM_STR` returned Real instead of String (E-25);
- format-string inference (`fmt_parser.rs`): terminates on every conversion character and reports it, so `%5d`/`%-8s`/`%08x` type their arguments correctly — previously only real conversions accepted flags/width (E-71);

- array inference: element-wise array `case`, array-literal function arguments (which previously bound nothing and silently returned 0), integer arrays typed Real, output-literal args accepted silently — all fixed (E-33); whole-array function arguments/outputs/returns (E-18, E-20, E-23); untyped function args default to Real instead of an ICE masquerading as an initializer bug (E-43);
- **`Type::base_type`** infinite loop on any array-type check (it looped on `self` instead of the cursor) — hung the compiler on simple type mismatches (E-14);
- string handling: ternary over strings (SELECT + string binary ops appended to the operator tables) (E-37);
- validation: loop bodies get their own context so the analog-operator restriction reports "loops" (LRM 4.5.1), not "conditions" (E-70); call-graph cycles among analog functions are clean errors naming the cycle instead of a recursive-inliner stack overflow (E-59); casex/casez restrictions (stray don't-care literals, `x` digits under `casez`, non-integer discriminants) (E-78); `domain discrete` with natures rejected per LRM 3.6.2.2 (E-50); parameter defaults exempt from their own ranges (the CMC "feature disabled" idiom — stock rejected diode_cmc, BSIM-CMG, PSP-HV and the HiSIM family at setup) while given values stay fully validated (E-56);
- contributions to port branches diagnosed (`ContributeToPortFlow`, with the declaration site labeled): `I(pb) <+ …` on a `branch (<p>) pb;` slipped through the write path unvalidated and panicked during lowering (E-84).

8. Lowering — `openvaf/hir_lower/`

`expr.rs`, `stmt.rs`, `ctx.rs`, `callbacks.rs`, `parameters.rs`, `state.rs`, `fmt.rs`, `lib.rs`:

- analog operators: `laplace_nd/np/zd/zp` via exact state-space realization (E-4) with complex pole/zero pairs per the LRM's (re, im) convention (E-31); `zi_*` via bilinear transform (E-6); `slew/transition` as a saturating tracking loop (E-6) with the LRM negative `max_neg_rate` convention honored (the sign defect made `slew` ignore its input and ramp away) (E-61) and a DC-singularity clamp via an `EnableIntegration` identity (E-47); `idtmod`'s modulo wrap moved off the reactive residual (it diverged) and an argument- index bug fixed (E-27); `idt` initial conditions survive into transient (E-28); the `idt` assert/reset forms with smooth reset dynamics (E-52); `limexp` kept stateless as a documented decision (E-13);
- **`$table_model`**: 1-D piecewise-linear lowered to differentiable MIR so autodiff yields the Jacobian for free (E-16); N-D multilinear as recursive 1-D (E-17, E-40); natural cubic splines with the precomputed moment matrix (E-22);
- events: `cross/above/timer` with persistent previous-value slots (E-8); OR lists folded with a select-based bool-or (a raw `ior` ICEs const-eval) (E-59); final-step gating and phase-list AND-ing (E-53);
- variable persistence: genuine per-instance `hidden_state` storage behind a two-pass MIR build (E-7), typed slots so integer state stopped crashing instruction selection (E-32);
- callbacks (`callbacks.rs`, `fmt.rs`): the `$display` family with the full `[flags][width][.precision]` surface and `%h/%b/%r` translation (E-71); file I/O and string formatting through a generalized `PrintDst` sink (E-11); the deterministic seed-and-salt RNG lowering (E-10); `$simparam$str` (E-25); simulation-control return flags (E-55); `$discontinuity` (E-24);
- operator fixes: unary ~ emitted arithmetic negate; the constant folder sign-extended >> where the runtime was unsigned — const and runtime disagreed (E-37);
- masked casex/casez comparisons (two `iand`s ahead of the existing integer equality) (E-78); array function-argument writeback (E-20) and array returns via a function-named var-array (E-23);
- named port branches (`branch (<p>) pb;` — LRM 3.7.2): a flow probe of a PortFlow-kind branch routes through the same `CurrentKind::Port` param as a direct `I(<p>)`, so E-29's defining equation covers both spellings; probing one used to hit `unreachable!()` (E-84);
- literal **`if`** conditions lower only the taken branch — a dead analog operator (`transition()` under `if ((0))`, the exact shape the `$port_connected` rewrite produces) survived const-folding as a detached-but-interned op whose state setup read optimized-away values and aborted codegen (E-

84).

9. MIR core and optimization — **openvaf/mir***

mir/ (instructions.rs + instructions/generated.rs, dfg.rs with dfg/phis.rs and dfg/uses.rs, dominators.rs, flowgraph.rs with flowgraph/transversal.rs, layout.rs, cursor.rs, lib.rs, builder/generated.rs), mir_opt/ (simplify_cfg.rs, simplify.rs, const_eval.rs, global_value_numbering.rs), mir_autodiff/ (builder.rs, lib.rs, live_derivatives.rs), mir_llvm/ (builder.rs, intrinsics.rs), mir_build/, mir_interpret/:

- three pre-existing general CFG bugs, all found via event-counter verification: a dangling-reference bug in unreachable-block removal, a multi-exit post-dominance bug in the dominator-tree builder, and a block-merge bug that could corrupt which block the DAE builder treated as the exit (E-8);
- the post-dominator sink-root pathology that let dead-code elimination delete op-dependent $fatal calls (side-effecting callbacks under op-dependent control now stay in eval; control dependence computed by arm-reachability) (E-55);
- split_tainted.rs tolerates layout-detached branch instructions (a branch whose condition const-folded has no CFG effect to taint; it used to unwrap and panic) (E-84);
- const-eval agreement with runtime semantics for shifts and the bitwise-not fix (E-37);
- autodiff: noise operators process before any ddt (a shared ddt evaluated between two noise operators silently dropped the second source), and the reactive coupling of a ddt-shaped noise wave is registered so small-signal pruning keeps it (E-54);
- **llvm.fabs.f64** registered in the LLVM intrinsics table — it never had been, and the signed noise-power convention needed it (E-42);
- new opcodes/instructions supporting the operator work (barrier/ integration-identity forms; MIR deliberately has no fabs, so lowering composes it) (E-47, E-52).

10. DAE construction — **openvaf/sim_back/**

dae.rs + dae/builder.rs, topology* (incl. lineralize.rs, small_signal_network.rs), noise.rs, context.rs, lib.rs:

- implicit equations for indirect branch assignment (one new unknown
  - residual per statement) (E-2) and the absdelay synthetic two-node form (E-1) — with equation indices kept stable by mapping dead/collapsed unknowns to zero-contribution placeholders instead of renumbering;
- port-flow probes **I(<port>)**: a TODO stub returning 0 became a synthesized DAE unknown mirroring the node's KCL residual (E-29);
- probe-only branches: probing a never-contributed branch read 0 and an open circuit — a 0 V-source pass materializes them, enabling ideal ammeters and flow-only signal-flow disciplines (E-36);
- noise factors (noise.rs): equation-path noise was silently lost (never attached to the DAE); factors generalize to re + jω·im with no extra matrix unknowns (E-54); same-named source grouping metadata (E-42);
- idt reset-mode charge dynamics and conditional step bounding (E-52).

11. OSDI code generation — **openvaf/osdi/**

lib.rs, metadata.rs + metadata/osdi_0_4.rs, inst_data.rs, eval.rs, load.rs, setup.rs, compilation_unit.rs, ndatable.rs, stdlib.c, build.rs, reference headers:

- descriptor emission for every ABI evolution (see the ngspice report's ABI table): the absdelay/last_crossing info arrays (E-1, E-6), OsdiNode.nodeset (E-45), the ac_stim source array + load_ac_stim (E-51), stride-2 signed noise pairs in load_noise (E-54) — the version tuple lives in osdi/src/lib.rs;

- **`eval.rs`**: final-step flag gating (E-53); simulation-control return flags (E-55); the `ac_stim` large-signal value (was an `unreachable!()` panic) (E-26);
- **`inst_data.rs`**: absdelay slot layout (E-1); hidden-state slots typed by the variable (`ty_f64` was hardcoded — integer state fed f64 loads into integer MIR ops and aborted instruction selection) (E-32);
- **`compilation_unit.rs`** (print codegen): the `%b` segfault — the binary-formatted string was remembered for `free()` but never pushed to the `snprintf` argument list, so the matching `%s` consumed a garbage pointer (E-71); the print-callback machinery with its shadowed-`fun` and volatile-table IPO traps (E-11);
- **`stdlib.c`**: the file-descriptor table behind `$fopen`/`$fdisplay`/… (E-11); the splitmix64 `osdi_rng_*` runtime (E-10); a `simparam_str` loop/return bug (E-25);
- **`ndatable.rs`**: `ddt`/`idt` natures resolved through `resolve_nature_index` instead of panicking on derived natures (E-39);
- **`setup.rs`**: range validation with default-exemption (E-56); a 64-line abandoned unsafe experiment (`VoidAbortCallback`) deleted in the warning cleanup (E-66);
- **`build.rs`**: the copied-target stdlib compilation trap fixed (E-10).

12. Driver, libraries, and infrastructure

- `openvaf-driver/src/main.rs`: hard errors now exit non-zero — the error arm printed the failure but fell through to a success exit, so every elaboration failure looked like a successful compile to shell scripts (a quirk first noticed during E-58's work) (E-84);
- `openvaf-driver/src/crash_report.rs`, `linker/src/lib.rs`, `lib/bforest/` (`map.rs`, `pool.rs`, `set.rs`), `lib/typed_indexmap/` (`set.rs`): the zero-warning cleanup (44 → 0 across 13 crates: lifetime annotations, dead code, nightly cfgs, an unused LLVM-prefix probe) (E-66);
- `basedb/` (`ast_id_map.rs`, `lints.rs`, `lib.rs`, `diagnostics/preprocessor_error.rs`, `diagnostics/syntax_error.rs`): AST-id plumbing for elaboration-created items (the `AstId::from_erased` placeholder used by array returns; E-23), preprocessor/syntax diagnostic rendering for the new errors;
- `.llvm-version` pins the LLVM 18 toolchain expectation; the `hir` and `preprocessor` `Cargo.toml`s carry the dependency additions their new code needed.

13. The test suite — **`openvaf/openvaf/tests/`**, **`lib/mini_harness/`**, **`test_data/`**, **`integration_tests/`**

The fork's own integration suite over real compact models had never run in this project: its OSDI loader was frozen at ABI 0.4, its optional VACASK legs panicked on a never-initialized submodule, and the tests hide behind `RUN_DEV_TESTS=1`. Fixed test-side only (E-68):

- `tests/load/osdi_0_4.rs` + `load/mod.rs`: loader structs synced to ABI 0.7; `mock_sim/mod.rs`: stride-2 noise convention; `mini_harness`: missing directories skip with a note instead of panicking;
- VACASK removed entirely (`.gitmodules`, the submodule, 34 harness legs and their stale snapshots): AGPL-3.0 cannot be vendored here;
- `test_data/` (95 files as a class): new fixtures for new features and snapshots regenerated whenever descriptor contents legitimately changed — every regeneration reviewed (the diffs are the project's own features: `flow(<port>)` unknowns, probe-only branches, implicit-equation nodes); `integration_tests/*/*.osdi` regenerated at each ABI bump.

---

14. Per-enhancement index (compiler-touching enhancements)

Enhancement	Pipeline areas	One line
E-1	sim_back, osdi	`absdelay()` synthetic-node DAE + descriptor arrays
E-2	parser→hir_lower, sim_back	indirect branch assignment (implicit equations)
E-3	parser, hir_def	vectored (bus) nets with bit-selects
E-4	hir_lower, syntax, parser	`laplace_*` state-space + array-literal parsing
E-5	elaborate.rs (new), parser	module instantiation via compile-time flattening
E-6	preprocessor, parser, hir_lower, osdi	directives, shifts, slew/transition, zi_*, last_crossing
E-7	hir_lower (state), osdi	`@(initial_step)` gating + variable persistence
E-8	parser, elaborate, hir_lower, mir/mir_opt	generate for + events; three general CFG bug fixes
E-9	hir_def, hir_lower, sim_back	noise tables; localparam/ground/strings; repeat/disable
E-10	hir_lower, osdi stdlib	`$random`/`$dist_*` deterministic RNG
E-11	hir_lower, osdi	file I/O + string functions (PrintDst)
E-12	hir_def, hir_lower	last unsupported builtins as LRM fallbacks
E-13	hir_lower	`limexp` kept stateless (documented decision)
E-14	hir_def, hir_ty, hir_lower	array literals/params/dynamic indexing; base_type hang
E-15	hir_def, hir_ty, hir_lower	N-D arrays
E-16/17/22/40	hir_ty, hir_lower	`$table_model`: 1-D, N-D multilinear, cubic spline, varargs sigs
E-18/20/23/33	hir_ty, hir_lower, basedb	arrays in analog functions: in/out/return/literals + array case
E-19	tokens→hir_lower	`do … while`
E-21/44	parser, elaborate, hir_def, sim_back	paramset + hidden system parameters
E-24/55	hir_lower, mir, osdi	`$discontinuity`; `$finish`/`$stop`/`$fatal` honored
E-25	hir_ty, osdi stdlib	`$simparam$str`
E-26/51	osdi, sim_back	`ac_stim`: crash fix, then full AC-RHS injection
E-27/28/52	hir_lower, sim_back	the `idt` family: modulo, IC, assert/reset
E-29/36	sim_back	port-flow probes; probe-only branches
E-30	hir_ty, hir_lower	variadic `analysis()`
E-31	hir_lower	complex poles/zeros in root forms
E-32	osdi inst_data	integer persistent state (crash fix)
E-34	tokens→hir_lower	`{…}` concat + `{n{…}}` replication
E-35	lexer	EOF comment hang
E-37/38	hir_lower, mir_opt, parser	operator + precedence audits (~, >> fold, `%` level)
E-39	parser, hir_ty, osdi	derived natures unwired at the node-kind boundary
E-41/49/58/59	elaborate	implicit nets; hierarchical names; defparam; port concat + OR lists

Enhancement	Pipeline areas	One line
E-42/54	sim_back noise, mir_autodiff, mir_llvm, osdi	correlated + node-free complex noise factors
E-43	hir_def (+diagnostics), hir_ty	initializers completed; arity diagnostics
E-45	osdi, hir_def, elaborate	nodesets + nature-attribute access (ABI 0.5)
E-46/48	lexer, syntax	escaped ids, based literals, single-pass unescaper
E-47	preprocessor, hir_ty, hir_lower	`` `default_transition `` + transition fixes
E-50	hir_ty validation	domain-binding validation
E-53	hir_lower, osdi eval	`@(final_step)` + phase lists
E-56	hir_ty, osdi setup	parameter defaults exempt from ranges
E-59	tokens→hir_lower, hir_ty	`$realtime`; OR lists; recursion diagnostics
E-61	hir_lower	`slew` sign fix (operator-args audit)
E-65	preprocessor	macro-recursion guard
E-66	13 crates	zero-warning cleanup (44 → 0)
E-67	tokens, parser, syntax, elaborate	generate audit: genvar fix, nesting, if/case
E-68	tests, mini_harness	integration suite enabled; VACASK removed
E-70	hir_ty validation	loop-context diagnostics
E-71	hir_ty fmt, hir_lower fmt, osdi	display-format surface + `%b` segfault
E-78	lexer→hir_lower, hir_ty	`casex`/`casez` don't-care masks
E-84	parser, elaborate, hir_ty, hir_lower, mir_opt, driver	LRM example sweep: 6 defect fixes (port-branch panic, parser robustness, silent undefined modules, `$port_connected` on open ports, dead-op codegen, exit codes)

Enhancements not listed (57, 60, 62–64, 69, 72–77, 79–83) changed no compiler sources — they were validation suites, documentation, benchmark tooling, or ngspice-side work (see the ngspice report).